

# IMAT MOCK EXAM 5 — "The Discriminator"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam

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## Section A - Logical Reasoning & Reading (Q1-9)

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1. "In oxidative phosphorylation, the proton gradient across the inner mitochondrial membrane is coupled to ATP synthesis. If the membrane is made permeable to protons by an uncoupler such as dinitrophenol, the gradient dissipates, electron transport accelerates, but ATP synthesis falls. Heat is produced instead."

Which is the strongest inference?

- A) Uncouplers raise ATP yield.
- B) The proton gradient is independent of electron transport.
- C) Uncoupling disconnects electron transport from ATP synthesis.
- D) Without a proton gradient, ATP synthesis rises.
- E) Uncoupling stops electron transport.

2. "A population bottleneck occurs when a population's size is drastically reduced for at least one generation. The genetic diversity of the surviving population is typically much lower than that of the original, because alleles are lost by chance. After the bottleneck, even rare alleles that remain can rise in frequency by genetic drift."

Which follows?

- A) Genetic drift stops after a bottleneck.
- B) Bottlenecks have no evolutionary consequence.
- C) Bottlenecks reduce genetic diversity regardless of which alleles happen to survive.
- D) After a bottleneck, the surviving diversity depends partly on which alleles randomly survived.
- E) Bottlenecks always raise genetic diversity.

3. (General knowledge) The Italian physician and educator who developed an eponymous educational method emphasising self-directed activity in a prepared environment, and was nominated for the Nobel Peace Prize multiple times, was:

- A) Rita Levi-Montalcini
- B) Maria Montessori
- C) Elena Lucrezia Cornaro Piscopia
- D) Sofia Vanni
- E) Nilde Iotti

4. Five people — F, G, H, I, J — are in a queue. F is immediately behind G. H is somewhere in front of G. I is not at either end. J is at the back. Which must be true?

- A) G is at the front.
- B) F is third.
- C) I is between H and G.
- D) H is at the front.
- E) F is immediately in front of J.

5. "All metals conduct electricity. X conducts electricity." From these, we can conclude:

- A) X might or might not be a metal (affirming the consequent is invalid).
- B) X is a metal.
- C) No metal is X.
- D) X is not a metal.
- E) X is an insulator.

6. A number, when divided by 3, gives remainder 2; when divided by 5, gives remainder 3. The smallest positive such number is:

- A) 5   B) 7   C) 8   D) 11   E) 13

7. A clock reads 8:20. What is the angle between the hands?  
**A)** 0° **B)** 100° **C)** 120° **D)** 130° **E)** 140°
8. If the 5th of a month is a Tuesday, what day is the 27th of the same month?  
**A)** Tuesday **B)** Wednesday **C)** Thursday **D)** Saturday **E)** Friday
9. Complete the letter series: A, C, F, J, O, ?  
**A)** S **B)** T **C)** V **D)** W **E)** U

## **Section B - Biology (Q10-32)**

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10. In glycolysis, the committed (rate-limiting) step is catalysed by:  
**A)** Phosphofructokinase-1 (PFK-1)  
**B)** Aldolase  
**C)** Phosphoglycerate kinase  
**D)** Hexokinase  
**E)** Pyruvate kinase
11. In the electron transport chain, FADH<sub>2</sub> donates electrons to:  
**A)** Complex II (succinate dehydrogenase)  
**B)** Complex IV  
**C)** ATP synthase  
**D)** Complex III  
**E)** Complex I
12. The chemiosmotic synthesis of ATP in mitochondria is driven by the flow of protons through ATP synthase from:  
**A)** Cytosol to matrix  
**B)** Intermembrane space to matrix  
**C)** Matrix to intermembrane space  
**D)** Outer to inner membrane (no gradient)  
**E)** Matrix to cytosol
13. Under anaerobic conditions, yeast cells regenerate NAD<sup>+</sup> by reducing pyruvate to:  
**A)** Water  
**B)** Acetyl-CoA  
**C)** Ethanol and CO<sub>2</sub>  
**D)** Glucose  
**E)** Lactate
14. During muscle contraction, calcium ions released from the sarcoplasmic reticulum bind to:  
**A)** Myosin heads  
**B)** Actin directly  
**C)** The sarcolemma only  
**D)** ATP  
**E)** Troponin, causing tropomyosin to move and expose myosin-binding sites
15. The sliding filament theory states that during contraction:  
**A)** The Z-lines move apart.  
**B)** Actin filaments shorten.  
**C)** Myosin filaments shorten.  
**D)** The sarcomere shortens, but the filaments themselves do not change length (they slide past each other).  
**E)** Both filaments shorten in proportion.

- 16.** Slow-twitch (Type I) muscle fibres are specialised for:
- A)** Replacing damaged neurons
  - B)** Generating action potentials
  - C)** Storing calcium
  - D)** Rapid, powerful, short-duration contractions using anaerobic glycolysis
  - E)** Endurance activity using oxidative metabolism (many mitochondria, high myoglobin)
- 17.** Most CO<sub>2</sub> transported in the blood is carried:
- A)** Dissolved as CO<sub>2</sub> gas in plasma
  - B)** As bicarbonate ions (HCO<sub>3</sub><sup>-</sup>) in plasma
  - C)** Bound to albumin
  - D)** Within erythrocyte nuclei
  - E)** Bound to haemoglobin (carbaminohaemoglobin)
- 18.** The enzyme carbonic anhydrase, abundant in red blood cells, catalyses:
- A)** CO<sub>2</sub> + H<sub>2</sub>O ⇌ H<sub>2</sub>CO<sub>3</sub>
  - B)** Glycolysis
  - C)** The conversion of haemoglobin to oxyhaemoglobin
  - D)** Pyruvate → acetyl-CoA
  - E)** The clotting cascade
- 19.** In the nephron, the majority of filtered water and solutes is reabsorbed in the:
- A)** Loop of Henle (descending limb only)
  - B)** Distal convoluted tubule
  - C)** Collecting duct
  - D)** Proximal convoluted tubule (~65%)
  - E)** Renal pelvis
- 20.** Antidiuretic hormone (ADH) increases water reabsorption by inserting aquaporins into the:
- A)** Afferent arteriole
  - B)** Proximal tubule
  - C)** Collecting duct (principal cells)
  - D)** Glomerulus
  - E)** Loop of Henle (thin limb)
- 21.** Aldosterone, secreted by the adrenal cortex, acts on the kidney to increase reabsorption of:
- A)** Glucose
  - B)** Urea
  - C)** K<sup>+</sup>
  - D)** Albumin
  - E)** Na<sup>+</sup> (and water), and increase K<sup>+</sup>/H<sup>+</sup> excretion
- 22.** In nervous tissue, the myelin sheath is produced in the CNS by \_\_ and in the PNS by \_\_.
- A)** Oligodendrocytes / Schwann cells
  - B)** Ependymal cells / astrocytes
  - C)** Astrocytes / Schwann cells
  - D)** Microglia / satellite cells
  - E)** Neurons / oligodendrocytes
- 23.** During an action potential, the rapid depolarisation is due to:
- A)** K<sup>+</sup> efflux
  - B)** Ca<sup>2+</sup> influx at the soma
  - C)** Na<sup>+</sup> influx (voltage-gated Na<sup>+</sup> channels open)
  - D)** Cl<sup>-</sup> influx
  - E)** Active transport of Na<sup>+</sup> out

- 24.** A cross between two plants both with red flowers produces 3 red : 1 white offspring. This is consistent with:
- A)** Polygenic inheritance
  - B)** Complete dominance (red dominant, parents heterozygous)
  - C)** Sex-linkage
  - D)** Incomplete dominance
  - E)** Codominance
- 25.** A man with blood group A (genotype AO) and a woman with blood group B (genotype BO) can have children with which blood groups?
- A)** O only
  - B)** A only
  - C)** B only
  - D)** A, B, AB, and O
  - E)** AB only
- 26.** DNA polymerase synthesises new DNA in which direction?
- A)** 3' → 5' on the new strand
  - B)** From the C-terminus
  - C)** Without direction
  - D)** In both directions simultaneously on one strand
  - E)** 5' → 3' on the new strand
- 27.** The genetic code is described as "degenerate" because:
- A)** Codons overlap.
  - B)** One codon codes for multiple amino acids.
  - C)** The code is ambiguous (one codon → multiple amino acids).
  - D)** Multiple codons can code for the same amino acid.
  - E)** The code differs between species.
- 28.** A bacterial operon whose transcription is increased in the presence of a small molecule (e.g., lactose for the lac operon) is:
- A)** Inducible (the molecule is an inducer)
  - B)** Constitutive
  - C)** Repressible
  - D)** Always off
  - E)** Negatively regulated only
- 29.** Which cell type produces antibodies?
- A)** Macrophages
  - B)** Platelets
  - C)** B cells (plasma cells, the differentiated form)
  - D)** Mast cells
  - E)** T-helper cells
- 30.** In photosynthesis, the oxygen released comes from:
- A)** O<sub>2</sub> taken from the air
  - B)** H<sub>2</sub>O (water, split in the light reactions)
  - C)** Glucose
  - D)** CO<sub>2</sub>
  - E)** NADPH
- 31.** In a DNA double helix, 20% of bases are adenine. The percentage of guanine is:
- A)** 20%   **B)** 30%   **C)** 40%   **D)** 60%   **E)** 80%

32. A mutation that inserts a single nucleotide into a coding sequence (not in multiples of 3) is called a:
- A) Point mutation only
  - B) Nonsense mutation
  - C) Silent mutation
  - D) Missense mutation
  - E) Frameshift mutation

### Section C - Chemistry (Q33-47)

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33. How many grams of  $\text{CO}_2$  are produced when 0.5 mol of C reacts completely with  $\text{O}_2$ ? (C = 12, O = 16.)
- A) 4.4 g   B) 11 g   C) 22 g   D) 44 g   E) 88 g
34. Balance:  $\_\_\text{Fe} + \_\_\text{O}_2 \rightarrow \_\_\text{Fe}_2\text{O}_3$ . The coefficient of Fe is:
- A) 1   B) 2   C) 3   D) 4   E) 6
35. What volume of 0.1 M  $\text{H}_2\text{SO}_4$  is needed to neutralise 25 mL of 0.2 M NaOH? ( $\text{H}_2\text{SO}_4$  is diprotic.)
- A) 5 mL   B) 12.5 mL   C) 25 mL   D) 50 mL   E) 100 mL
36. The pH of a buffer containing 0.1 M acetic acid ( $\text{pK}_a = 4.76$ ) and 0.1 M sodium acetate is:
- A) 3.76   B) 4.76   C) 5.76   D) 7.0   E) 9.24
37. The oxidation number of oxygen in hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) is:
- A) -2   B) -1   C) 0   D) +1   E) +2
38. Which species is reduced in:  $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ ?
- A)  $\text{Cl}_2$  ( $\text{Cl}_2 \rightarrow 2\text{Cl}^-$ )
  - B) Na ( $\text{Na} \rightarrow \text{Na}^+$ )
  - C) Nothing
  - D) NaCl
  - E) Both Na and  $\text{Cl}_2$
39. An ideal gas occupies 2 L at 1 atm and 300 K. What is its volume at 2 atm and 300 K?
- A) 0.5 L   B) 1 L   C) 2 L   D) 4 L   E) 8 L
40. How many electrons can occupy the d-subshell (5 orbitals)?
- A) 2   B) 6   C) 8   D) 10   E) 14
41. Which element has the electron configuration  $1s^2 2s^2 2p^6 3s^1$ ?
- A) Na   B) K   C) Mg   D) Al   E) Ne
42. The geometry of  $\text{NH}_3$  (ammonia) is:
- A) Trigonal planar
  - B) Octahedral
  - C) Square planar
  - D) Linear
  - E) Tetrahedral (electron-pair geometry) / trigonal pyramidal (molecular shape)
43. Which compound is most soluble in water?
- A)  $\text{CH}_3\text{OH}$  (methanol)
  - B)  $\text{CCl}_4$
  - C)  $\text{CH}_4$  (methane)
  - D)  $\text{C}_6\text{H}_6$  (benzene)
  - E)  $\text{C}_2\text{H}_6$  (ethane)

44. An exothermic reaction has:  
A) No activation energy  
B) Products higher in energy than reactants  
C)  $\Delta H > 0$   
D)  $\Delta H = 0$   
E)  $\Delta H < 0$
45. The IUPAC name of  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$  is:  
A) Propane B) Ethane C) Methane D) Butane E) Pentane
46. For the equilibrium  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ ,  $\Delta H < 0$ . Increasing pressure at constant T will:  
A) Shift left  
B) Shift right  
C) Have no effect (equal moles of gas on both sides)  
D) Decrease the rate  
E) Increase  $\Delta H$
47. If the rate constant doubles when temperature increases by  $10^\circ\text{C}$ , and the rate at  $T_1$  is  $r$ , the rate at  $T_1 + 20^\circ\text{C}$  is approximately:  
A)  $2r$  B)  $3r$  C)  $4r$  D)  $8r$  E)  $10r$

### Section D - Mathematics (Q48-54)

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48. Solve:  $x^2 - 5x - 6 = 0$ .  
A) No real roots B)  $x = 1, -6$  C)  $x = 2, 3$  D)  $x = 5, 1$  E)  $x = -1, 6$
49. Solve:  $\log_3(x) = -2$ .  
A)  $x = 1/9$  B)  $x = 9$  C)  $x = -2$  D)  $x = -6$  E)  $x = 1/3$
50. Solve:  $(2x - 1)/(x + 1) = 3$  ( $x \neq -1$ ).  
A)  $x = 1$  B)  $x = -2$  C)  $x = 2$  D)  $x = -4$  E)  $x = 4$
51. Solve the inequality:  $x^2 - 5x + 6 > 0$ .  
A)  $2 < x < 3$  B)  $x > 2$  C)  $x < 2$  or  $x > 3$  D)  $x < 3$  E) All real  $x$
52. A circle has circumference  $12\pi$ . What is its area?  
A)  $6\pi$  B)  $12\pi$  C)  $24\pi$  D)  $36\pi$  E)  $144\pi$
53. In a right triangle, the side opposite the  $30^\circ$  angle has length 5. What is the length of the hypotenuse?  
A) 5 B)  $5\sqrt{2}$  C)  $5\sqrt{3}$  D) 10 E) 15
54. Two dice are rolled. What is the probability that the sum is 7?  
A)  $1/12$  B)  $1/9$  C)  $1/6$  D)  $5/36$  E)  $7/36$

### Section E - Physics (Q55-60)

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55. A car starts from rest and accelerates uniformly at  $3 \text{ m/s}^2$ . How far does it travel in 4 s?  
A) 12 m B) 24 m C) 36 m D) 48 m E) 72 m
56. A  $6 \Omega$  and a  $3 \Omega$  resistor are connected in parallel across a 12 V battery. What is the total current from the battery?  
A) 1 A B) 2 A C) 3 A D) 4 A E) 6 A
57. A 100 W bulb is operated on a 200 V supply for 10 s. How much electrical energy is used?  
A) 20,000 J B) 100 J C) 200 J D) 1000 J E) 2000 J

**58.** A 2 kg ball moving at 5 m/s strikes a wall and rebounds at 3 m/s. What is the magnitude of the change in momentum?

- A)** 2 kg·m/s   **B)** 4 kg·m/s   **C)** 8 kg·m/s   **D)** 10 kg·m/s   **E)** 16 kg·m/s

**59.** A satellite orbits Earth at a radius where gravitational acceleration is  $2 \text{ m/s}^2$ . If Earth's radius is  $R$  and the satellite is at  $3R$  from the centre, the surface gravitational acceleration is:

- A)**  $2 \text{ m/s}^2$    **B)**  $6 \text{ m/s}^2$    **C)**  $9 \text{ m/s}^2$    **D)**  $18 \text{ m/s}^2$    **E)**  $27 \text{ m/s}^2$

**60.** A 0.5 kg block of iron ( $c = 450 \text{ J/kg} \cdot ^\circ\text{C}$ ) is heated from  $20^\circ\text{C}$  to  $100^\circ\text{C}$ . How much heat is required?

- A)** 9,000 J   **B)** 18,000 J   **C)** 22,500 J   **D)** 45,000 J   **E)** 90,000 J

## Answer key

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This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | C | 2  | D | 3  | B | 4  | E | 5  | A |
| 6  | C | 7  | D | 8  | B | 9  | E | 10 | A |
| 11 | A | 12 | B | 13 | C | 14 | E | 15 | D |
| 16 | E | 17 | B | 18 | A | 19 | D | 20 | C |
| 21 | E | 22 | A | 23 | C | 24 | B | 25 | D |
| 26 | E | 27 | D | 28 | A | 29 | C | 30 | B |
| 31 | B | 32 | E | 33 | C | 34 | D | 35 | C |
| 36 | B | 37 | B | 38 | A | 39 | B | 40 | D |
| 41 | A | 42 | E | 43 | A | 44 | E | 45 | D |
| 46 | C | 47 | C | 48 | E | 49 | A | 50 | D |
| 51 | C | 52 | D | 53 | D | 54 | C | 55 | B |
| 56 | E | 57 | D | 58 | E | 59 | D | 60 | B |